Problem Set 5

Due February 20, 2026 at 11:59 PM Pacific

Turn in your problem set by Gradescope. Include any code you write in your solutions as well. Mark down how much time you spent on the homework set.

1. [15 points] **State and Parameter Estimation:** The EKF can be used to estimate unknown parameters of a dynamical system, while also estimating its state. Consider the roll mode of a fixed-wing aircraft, with the model $\dot{p} = -ap + bu$, where p is roll rate, u is an aileron input, and a, b are unknown (fixed) model parameters. You receive a measurement $y(t) = p(t)$ from the IMU.

   (a) Formulate a discrete time approximation of this system $x_{t+1} = f(x_t, u_t) + w_t$, $y_t = g(x_t) + v_t$ with the state $x = [p \quad a \quad b]^T$.

   (b) Find the Jacobians A_t and C_t for the EKF.

   (c) Simulate the system with true parameters $a = 0.5$, $b = 2$, with an input $u_t = \sin(t)$, with $p(0) = 5$, and with additive process noise covariance $Q = \text{diag}([0.1 \quad 0 \quad 0])$ and measurement noise covariance $R = 0.1$. Implement an EKF to estimate the parameters a and b, as well as p. Use the noise covariances $Q = \text{diag}([0.1 \quad 0.01 \quad 0.01])$, $R = 0.1$ in your EKF. Start with $\Sigma_0 = 10I_3$ and $\mu_0 = 0$. Use $dt = 0.1s$, and simulate for 100 seconds.
   Remark: Think about what this result represents. You only have a model of your system, not an exact specification. But this method lets you estimate from measurements, in real-time, not just your real state but also the parameters of the system!

2. [15 points] **Kalman Filter as an Optimization Problem**: In class we claimed that the Kalman Filter update equations can be derived as the analytical solution to a least squares optimization problem. Let's prove the claim. Consider the Maximum A Posteriori (MAP) optimization problem we saw in class, $x_t^{\text{MAP}} = \arg\min_{x_t} J(x_t)$. With a linear measurement model $Y_t = C_t X_t + V_t$, $V_t \sim \mathcal{N}(0, R_t)$, and Gaussian prediction distribution $p(x_t \mid y_{1:t-1}) = \mathcal{N}(\mu_{t|t-1}, \Sigma_{t|t-1})$, we saw that the MAP objective function becomes a convex quadratic in x_t,

$$J(x_t) = \frac{1}{2}\|y_t - C_t x_t\|^2_{R_t^{-1}} + \frac{1}{2}\|x_t - \mu_{t|t-1}\|^2_{\Sigma^{-1}_{t|t-1}}.$$

   (a) Analytically find x_t^{MAP} by setting the gradient of the objective function to zero $\frac{\partial J(x_t)}{\partial x_t}\,|_{x_t = x_t^{\text{MAP}}} = 0$ and solving for x_t^{MAP}. Verify that $\mu_{t|t} = x_t^{\text{MAP}}$ by showing that your expression matches the mean update equation from the Kalman filter.

   (b) Analytically find the Hessian, $H = \frac{\partial^2 J(x_t)}{\partial x_t^2}\,|_{x_t = x_t^{\text{MAP}}}$. Show that $\Sigma_{t|t} = H^{-1}$ by verifying that the inverse of your expression for H matches the covariance update equation from the Kalman Filter.

   Hint: the matrix inverse lemmas that we have seen in previous problem sets will come in handy for this problem!

3. [20 points] **EKF SLAM**: Consider again the differential drive robot model,

$$\begin{aligned} P^x_{t+1} &= P^x_t + \delta t\, s_t \cos\Theta_t + W^x_t \\ P^y_{t+1} &= P^y_t + \delta t\, s_t \sin\Theta_t + W^y_t \\ \Theta_{t+1} &= \Theta_t + \delta t\, \phi_t + W^\theta_t \end{aligned}$$

   where $P_t = [P^x_t \quad P^y_t]^T$ is the robot position relative to a feature point, Θ_t is its heading angle, s_t its speed input, ϕ_t its rotation rate input, and W_t is the process noise distributed as $W_t \sim \mathcal{N}(0, Q)$ with $Q = 0.1\delta t I_3$. Suppose you have four features at positions $m_1 = [0 \quad 0]^T$, $m_2 = [10 \quad 0]^T$, $m_3 = [10 \quad 10]^T$, and $m_4 = [0 \quad 10]^T$. Now suppose the robot takes relative position measurements to each of the features (in the robot's local frame), with measurement model,

$$Y^j_t = \begin{bmatrix} \cos(\Theta_t) & \sin(\Theta_t) \\ -\sin(\Theta_t) & \cos(\Theta_t) \end{bmatrix} (m_j - P_t) + V^j_t,$$

as we saw in class. The measurements have an associated measurement uncertainty of $V_t^j \sim \mathcal{N}(0, R)$, where $R = 0.25I_2$. Also suppose that the robot pose is initially distributed $\mathcal{N}([1m\ 1m\ 0rad]^T, I_3)$ is commanded with a known control sequence $s_t = 0.5$ and $\phi_t = \sin(2t)$rad/s. You can assume that the ground truth trajectory also starts at $[1m\ 1m\ 0rad]^T$. Use $\delta t = 0.5s$ and run for 100 timesteps (i.e., $50s$).

We highly recommend using the starter code from Problem Set 4 Q4, and editing it as necessary for this problem.

(a) Calculate the measurement Jacobian C_t^b for the measurement model.

(b) Implement an EKF for the robot to localize itself with these measurements given known feature locations.

(c) Implement an EKF to map the features given known robot state trajectory. Use a *noiseless* ground-truth trajectory created by the given inputs as your reference. Start your initial mean positions at the origin, with a large covariance in the position ($\Sigma_0 = 10^5 I$)

(d) Combine the two items above to jointly estimate the robot pose and feature locations. This is EKF SLAM, the original SLAM solution.

For each of parts (b), (c), and (d), provide a plot with

- the mean estimate of the state (whether that is the robot's location, the feature positions, or both),
- the true trajectory and true positions of the features,
- the covariance ellipses around the mean position estimates.
- On a separate plot, include the Θ time series estimate, the true Θ trajectory, and the 95% confidence bounds around the estimate.

1. [15 points] **State and Parameter Estimation:** The EKF can be used to estimate unknown parameters of a dynamical system, while also estimating its state. Consider the roll mode of a fixed-wing aircraft, with the model $\dot{p} = -ap + bu$, where p is roll rate, u is an aileron input, and a, b are unknown (fixed) model parameters. You receive a measurement $y(t) = p(t)$ from the IMU.

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   Remark: Think about what this result represents. You only have a model of your system, not an exact specification. But this method lets you estimate from measurements, in real-time, not just your real state but also the parameters of the system!

a)

$f(x_t, u_t)$:
$p, a, b = x_t$
$\dot{p} = -ap + bu_t$
$\dot{x} = [\dot{p} \; 0 \; 0]$
return $x_t + \dot{x} \cdot dt$

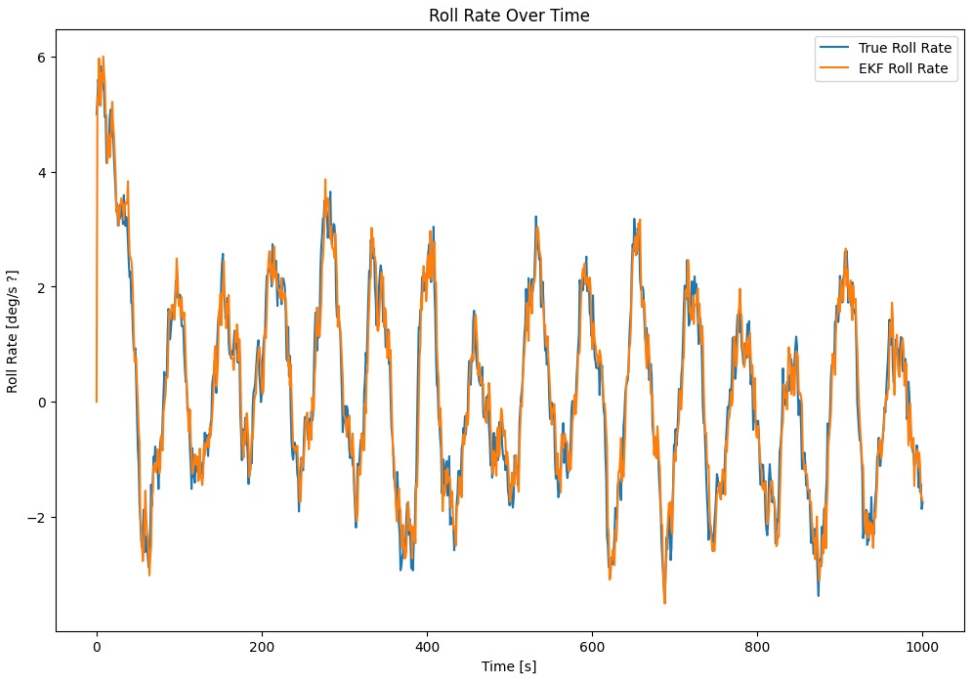
$g(x_t)$:
$p, a, b = x_t$
return p

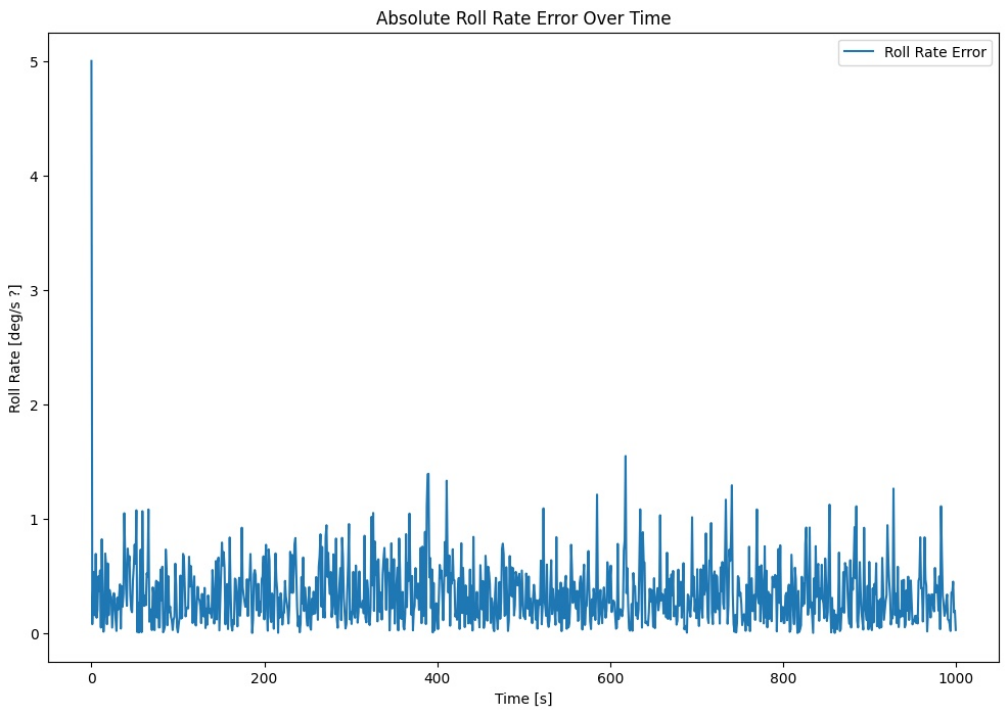
b)

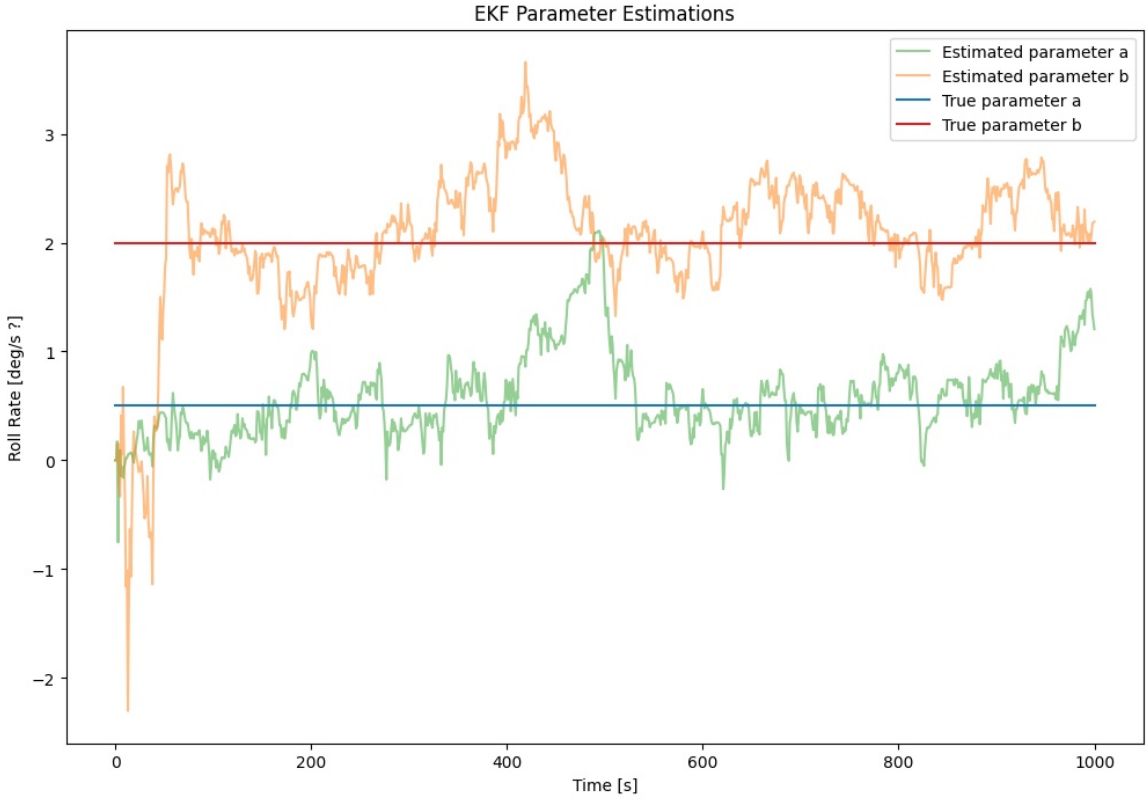
$$A = dt \begin{bmatrix} -a & -p & u \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

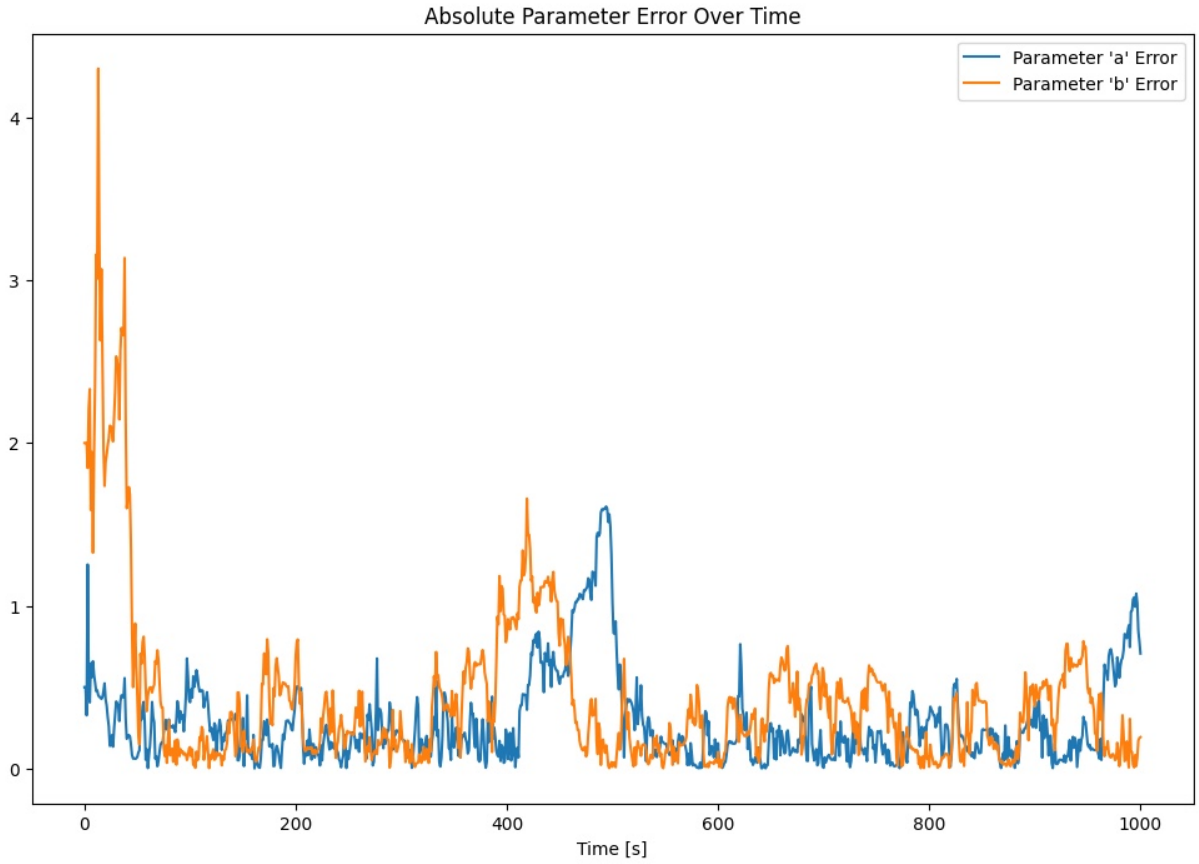
$$C = [1 \; 0 \; 0]$$

c)






EKF Parameter Estimations
Estimated parameter a
Estimated parameter b
True parameter a
True parameter b
Roll Rate [deg/s ?]
Time [s]
3
2
1
0
−1
−2
0
200
400
600
800
1000


Absolute Parameter Error Over Time
Parameter 'a' Error
Parameter 'b' Error
Time [s]
4
3
2
1
0
0
200
400
600
800
1000

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quadratic form derivative!

$$\frac{d}{dx}(x^T A x) = x^T(A + A^T)$$

matrix derivative

$$\frac{\partial A(y)}{\partial x} = \frac{\partial y}{\partial x}\frac{\partial A}{\partial y}$$

$$\|y_t - C_t x_t\|^2_{R_t^{-1}} = (y_t - C_t x_t)^T R_t^{-1}(y_t - C_t x_t)$$

$$\frac{\partial f}{\partial x} = (\nabla_x f)^T$$

$$\nabla_x J(x_t) = \frac{1}{2}\nabla_x \|y_t - C_t x_t\|^2_{R_t^{-1}} + \frac{1}{2}\|x_t - \mu_{t|t-1}\|^2_{\Sigma^{-1}_{t|t-1}}$$

(1)

$$\frac{\partial}{\partial x}\|y_t - C_t x_t\|^2_{R_t^{-1}} = \left[(y_t - C_t x_t)^T (R_t^{-1} + R_t^{-1})\right]\frac{\partial}{\partial x}(y_t - C_t x_t)$$

(R_t^{-1} symmetric)

$$= 2(y_t - C_t x_t)^T R_t^{-1}(-C_t) \rightarrow \nabla_x f_1 = \frac{\partial f_1}{\partial x}^T = -2C_t^T R_t^{-1}(y_t - C_t x_t)$$

(2)

$$\frac{\partial}{\partial x}\|x_t - \mu_{t|t-1}\|^2_{\Sigma^{-1}_{t|t-1}} = \left[(x_t - \mu_{t|t-1})^T(\Sigma^{-1}_{t|t-1} + \Sigma^{-1}_{t|t-1})\right]\frac{\partial}{\partial x}(x_t - \mu_{t|t-1})$$

$$= 2(x_t - \mu_{t|t-1})^T \Sigma^{-1}_{t|t-1} \qquad \nabla_x f_2 = \frac{\partial f_2}{\partial x}^T = 2\Sigma^{-1}_{t|t-1}(x - \mu_{t|t-1})$$

$$\therefore \quad \nabla_x J(x_t) = -C_t^T R_t^{-1}(y_t - C_t x_t) + \Sigma^{-1}_{t|t-1}(x - \mu_{t|t-1}) = 0$$

$$-C_t^T R_t^{-1}(y_t - C_t x_t) + \Sigma^{-1}_{t|t-1}(x - \mu_{t|t-1}) = 0$$

$$C_t^T R_t^{-1}(y_t - C_t x_t) = \Sigma^{-1}_{t|t-1}(x - \mu_{t|t-1})$$

$$C_t^T R_t^{-1} y_t - C_t^T R_t^{-1} C_t x_t = \Sigma^{-1}_{t|t-1} x_t - \Sigma^{-1}_{t|t-1}\mu_{t|t-1}$$

$$(C_t^T R_t^{-1} C_t + \Sigma^{-1}_{t|t-1}) x_t = (C_t^T R_t^{-1} y_t + \Sigma^{-1}_{t|t-1}\mu_{t|t-1})$$

$$x_t = (C_t^T R_t^{-1} C_t + \Sigma^{-1}_{t|t-1})^{-1}(C_t^T R_t^{-1} y_t + \Sigma^{-1}_{t|t-1}\mu_{t|t-1})$$

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$$C_t^T R_t^{-1} y_t + \Sigma^{-1}_{t|t-1}\mu_{t|t-1} + C_t^T R_t C_t \mu_{t|t-1} - C_t^T R_t C_t \mu_{t|t-1}$$

$$(C_t^T R_t^{-1} C_t + \Sigma_{t|t-1})\mu_{t|t-1} + C_t^T R_t^{-1}(y_t - C_t \mu_{t|t-1})$$

$$x_t = (C_t^T R_t^{-1} C_t + \Sigma^{-1}_{t|t-1})^{-1}\left[(C_t^T R_t^{-1} C_t + \Sigma_{t|t-1})\mu_{t|t-1} + C_t^T R_t^{-1}(y_t - C_t \mu_{t|t-1})\right]$$

$$x_t = \mu_{t|t-1} + \left(C_t^T R_t^{-1} C_t + \Sigma_{t|t-1}^{-1}\right)^{-1} C_t^T R_t^{-1}(y_t - C_t \mu_{t|t-1})$$

prev. HW

$$(A+UCV)^{-1}UC = A^{-1}U(C^{-1}+VA^{-1}U)^{-1}.$$

$$x_t = \mu_{t|t-1} + \Sigma_{t|t-1} C_t^T \left(R_t + C_t \Sigma_{t|t-1} C_t^T\right)^{-1}(y_t - C_t \mu_{t|t-1})$$

same as $\mu_{t|t}$ in KF update

$$\nabla_x J(x_t) = -C_t^T R_t^{-1}(y_t - C_t x_t) + \Sigma_{t|t-1}^{-1}(x - \mu_{t|t-1})$$

$$\nabla_x^2 J(x_t) = C_t^T R_t^{-1} C_t + \Sigma_{t|t-1}^{-1} = H$$

$$H^{-1} = \left(C_t^T R_t^{-1} C_t + \Sigma_{t|t-1}^{-1}\right)^{-1}$$

prev HW

$$(A+UCV)^{-1} = A^{-1} - A^{-1}U(VA^{-1}U + C^{-1})^{-1}VA^{-1}$$

$$H^{-1} = \Sigma_{t|t-1} - \Sigma_{t|t-1} C_t^T \left(C_t \Sigma_{t|t-1} C_t^T + R_t\right)^{-1} C_t \Sigma_{t|t-1}$$

same as $\Sigma_{t|t}$ in KF update

3. [20 points] **EKF SLAM**: Consider again the differential drive robot model,

$$P_{t+1}^x = P_t^x + \delta t\, s_t \cos\Theta_t + W_t^x$$
$$P_{t+1}^y = P_t^y + \delta t\, s_t \sin\Theta_t + W_t^y$$
$$\Theta_{t+1} = \Theta_t + \delta t\, \phi_t + W_t^\theta$$

where $P_t = [P_t^x \quad P_t^y]^T$ is the robot position relative to a feature point, Θ_t is its heading angle, s_t its speed input, ϕ_t its rotation rate input, and W_t is the process noise distributed as $W_t \sim \mathcal{N}(0, Q)$ with $Q = 0.1\delta t I_3$. Suppose you have four features at positions $m_1 = [0 \quad 0]^T$, $m_2 = [10 \quad 0]^T$, $m_3 = [10 \quad 10]^T$, and $m_4 = [0 \quad 10]^T$. Now suppose the robot takes relative position measurements to each of the features (in the robot's local frame), with measurement model,

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a) $A = \begin{bmatrix} 1 & 0 & -\delta t \cdot s \cdot \sin(\theta) \\ 0 & 1 & \delta t \cdot s \cdot \cos(\theta) \\ 0 & 0 & 1 \end{bmatrix}$

$C = \begin{bmatrix} \frac{\partial y_x^1}{\partial x} & \frac{\partial y_x^1}{\partial y} & \frac{\partial y_x^1}{\partial \theta} \\ \vdots & \vdots & \vdots \end{bmatrix}$

$y^j = \begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} (m_j - p_t)$

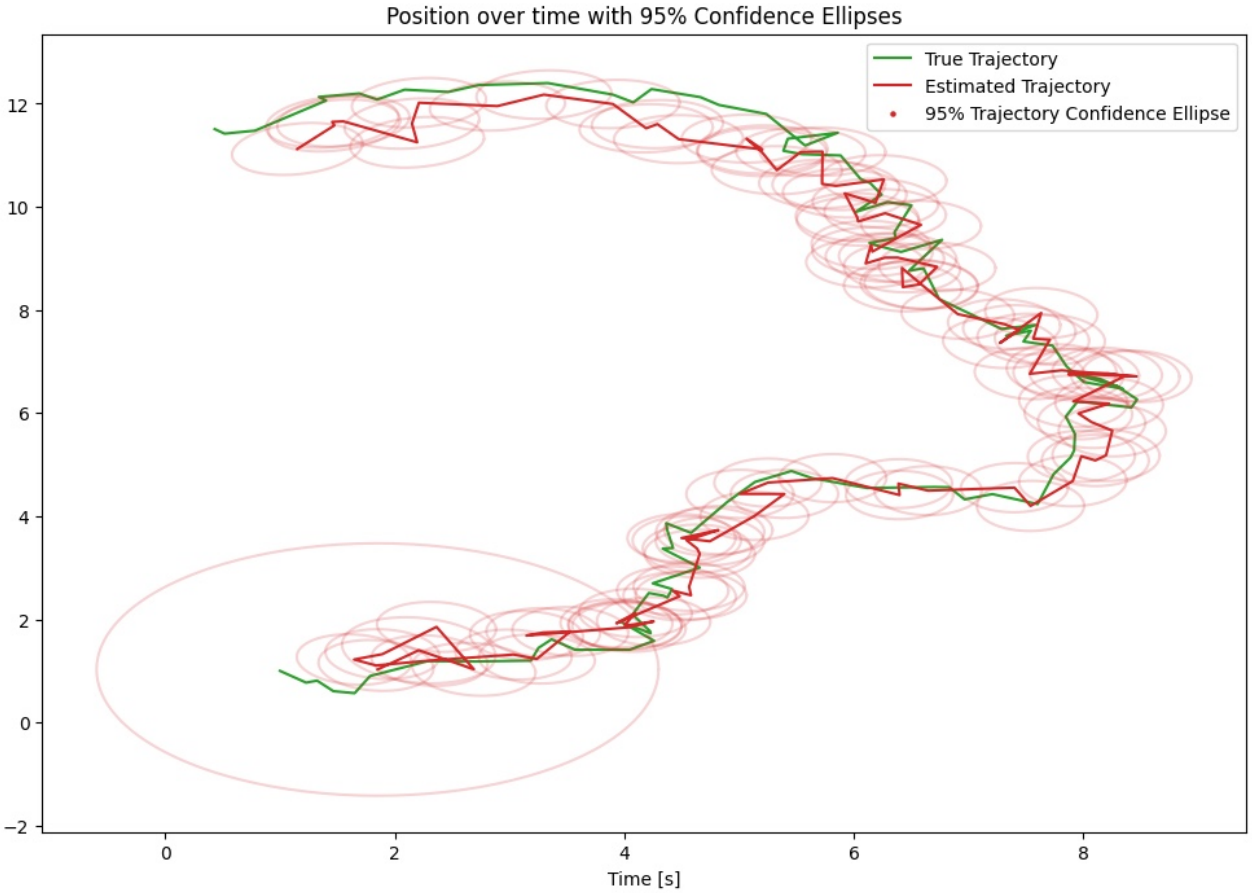
$\frac{\partial y^j}{\partial x} = \begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} \frac{\partial}{\partial x}(m_j - p_t) = -\begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} \frac{\partial p_t}{\partial x} = -\begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -c\theta \\ s\theta \end{bmatrix}$

$\frac{\partial y^j}{\partial y} = \begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} \frac{\partial}{\partial y}(m_j - p_t) = -\begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -s\theta \\ -c\theta \end{bmatrix}$

$\frac{\partial y^j}{\partial \theta} = \begin{bmatrix} -s\theta & c\theta \\ -c\theta & -s\theta \end{bmatrix} (m_j - p_t) = \begin{bmatrix} -s\theta (m_{j,x} - x) + c\theta (m_{j,y} - y) \\ -c\theta (m_{j,x} - x) - s\theta (m_{j,y} - y) \end{bmatrix}$

$C_t = \begin{bmatrix} -c\theta & -s\theta & -s\theta (m_{1,x} - x) + c\theta (m_{1,y} - y) \\ s\theta & c\theta & -c\theta (m_{1,x} - x) - s\theta (m_{1,y} - y) \\ -c\theta & -s\theta & -s\theta (m_{2,x} - x) + c\theta (m_{2,y} - y) \\ s\theta & c\theta & -c\theta (m_{2,x} - x) - s\theta (m_{2,y} - y) \\ -c\theta & -s\theta & -s\theta (m_{3,x} - x) + c\theta (m_{3,y} - y) \\ s\theta & c\theta & -c\theta (m_{3,x} - x) - s\theta (m_{3,y} - y) \\ -c\theta & -s\theta & -s\theta (m_{4,x} - x) + c\theta (m_{4,y} - y) \\ s\theta & c\theta & -c\theta (m_{4,x} - x) - s\theta (m_{4,y} - y) \end{bmatrix}$

5)



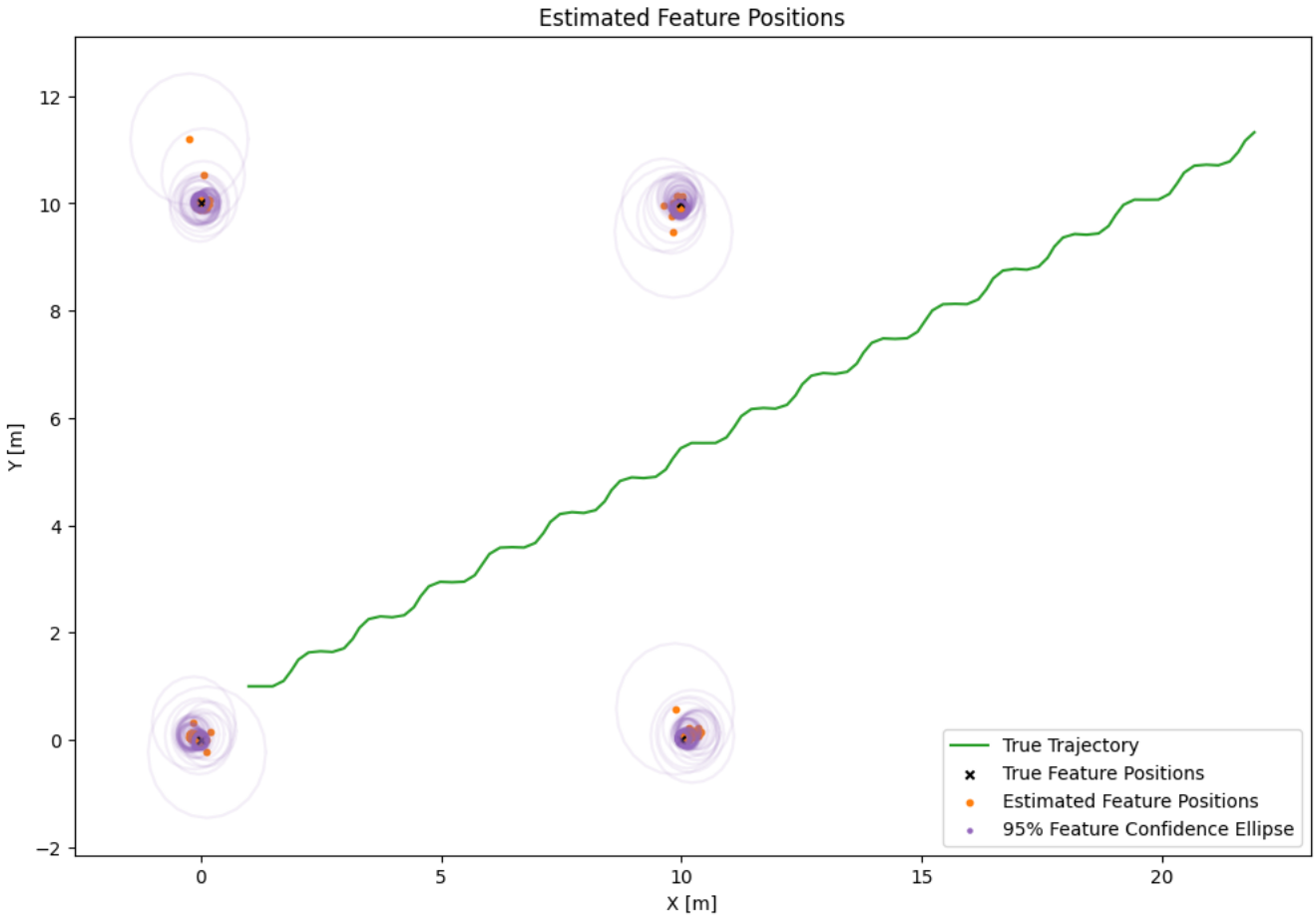
c)

$f[x,u] = 0 \quad A = I_6$ since features do not change position

$$y^j = \begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} (m_j - p_t) \qquad m_j - p_t = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix} y^j$$

$$m_j = p_t + \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix} y^j$$

$$\frac{\partial y^j}{\partial m_j} = \begin{bmatrix} c\theta & s\theta \\ -s\theta & c\theta \end{bmatrix} \qquad C = \begin{bmatrix} c\theta & s\theta & \cdots & \cdots & \cdots \\ -s\theta & c\theta & & & \\ \vdots & & c\theta & s\theta & \vdots \\ & & -s\theta & c\theta & \\ \cdots & \cdots & \cdots & c\theta & s\theta \\ \vdots & \vdots & \vdots & -s\theta & c\theta \end{bmatrix}$$



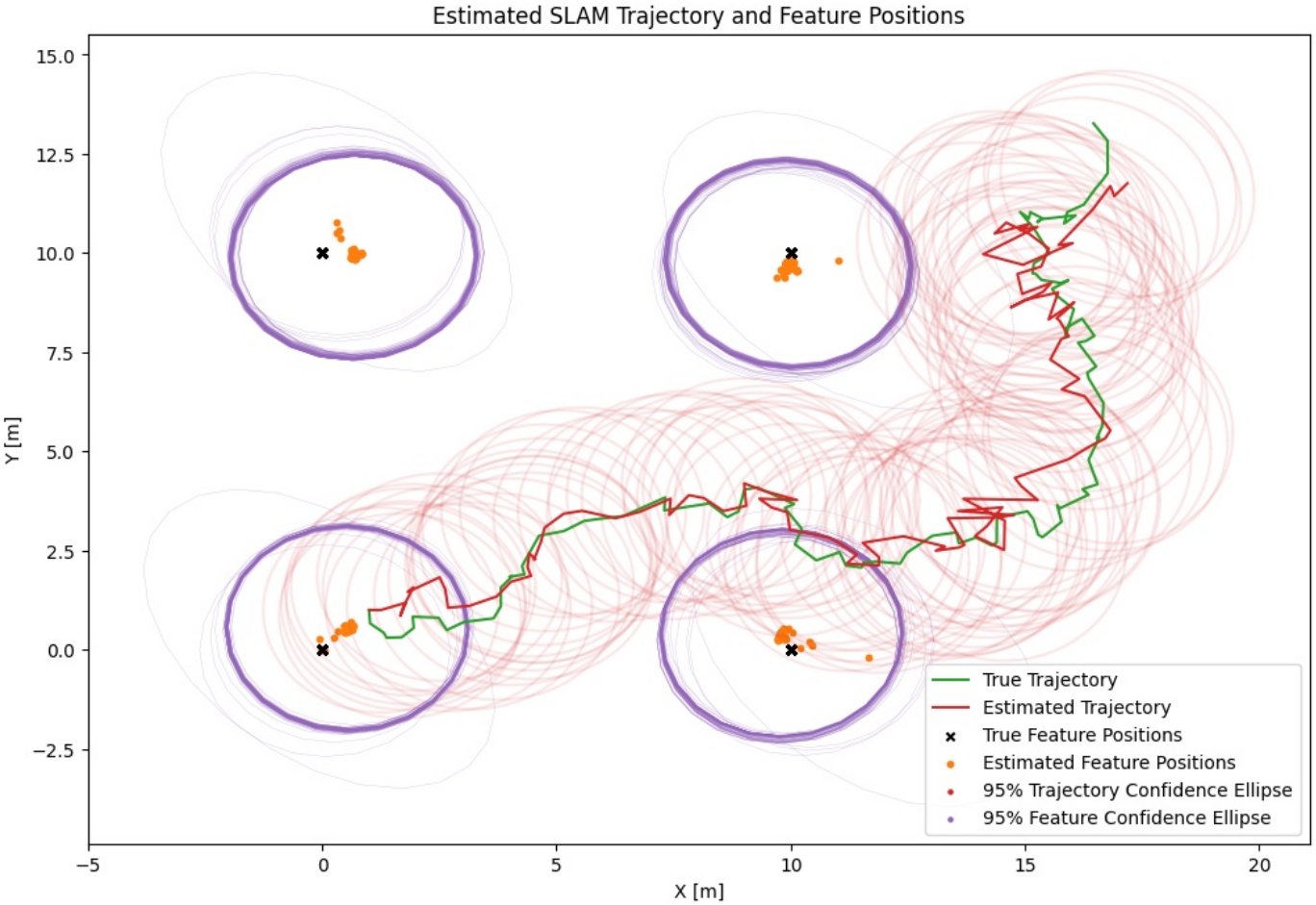
d) $X = (p_x, p_y, p_\theta, m_{x1}, m_{y1}, m_{x2}, m_{y2}, m_{x3}, m_{y3}, m_{x4}]^T = [x, m]^T$

$f(x, u, \delta t)$

$s, \phi = u$

$$x_{t+1} = \begin{bmatrix} p_{t+1} \\ m_{t+1} \end{bmatrix} = \begin{bmatrix} p_x + \delta t \cdot s_t \cdot \cos\theta \\ p_y + \delta t \cdot s_t \cdot \sin\theta \\ p_\theta + \delta t \cdot \phi \\ m_{ix} \\ m_{iy} \\ \vdots \end{bmatrix}$$

$$A = \begin{bmatrix} A_{robot} & 0 \\ 0 & A_{features} \end{bmatrix} \qquad C = \begin{bmatrix} C_{robot} & C_{features} \end{bmatrix}$$


Estimated SLAM Trajectory and Feature Positions
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
Y [m]
−5
0
5
10
15
20
X [m]
True Trajectory
Estimated Trajectory
True Feature Positions
Estimated Feature Positions
95% Trajectory Confidence Ellipse
95% Feature Confidence Ellipse

wow!